Mismatched Petz recovery for the Leung code under weak amplitude damping

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November 3, 2026

Abstract

The Petz recovery map is built from a model of the noise. What happens when this model is wrong? We study the four-qubit Leung code under amplitude damping with parameter γ, recovered by the Petz map built for an assumed parameter h. Numerically, the best choice is not $h = \gamma$ but a smaller value, which approaches $\gamma/2$ at weak noise. We prove that for $h = r\gamma$ with $r > 0$ fixed,

$$F(\gamma, r\gamma) = 1 - \left[\tfrac{7}{4} - \left(1 - \tfrac{1}{\sqrt{2}}\right) r(1-r)\right]\gamma^2 + O(\gamma^3),$$

so $r = \frac{1}{2}$ is optimal, and any underestimate of the noise ($0 < r < 1$) beats the matched map. The proof rests on two structural facts. First, 236 of the 256 traces entering the fidelity vanish identically. Second, the first-order terms cancel, because every Petz map with $h > 0$ corrects single decays exactly at leading order.

1 Introduction

The Petz map [1] is a universal recovery channel. Given a noise channel and a reference state, it provides an explicit decoder whose fidelity is close to the optimal one [2]. Its weak point is that it needs the noise channel itself. In practice a damping rate is estimated, not known, so the decoder is built for an assumed parameter h while the true parameter is γ. How much fidelity does a wrong guess cost?

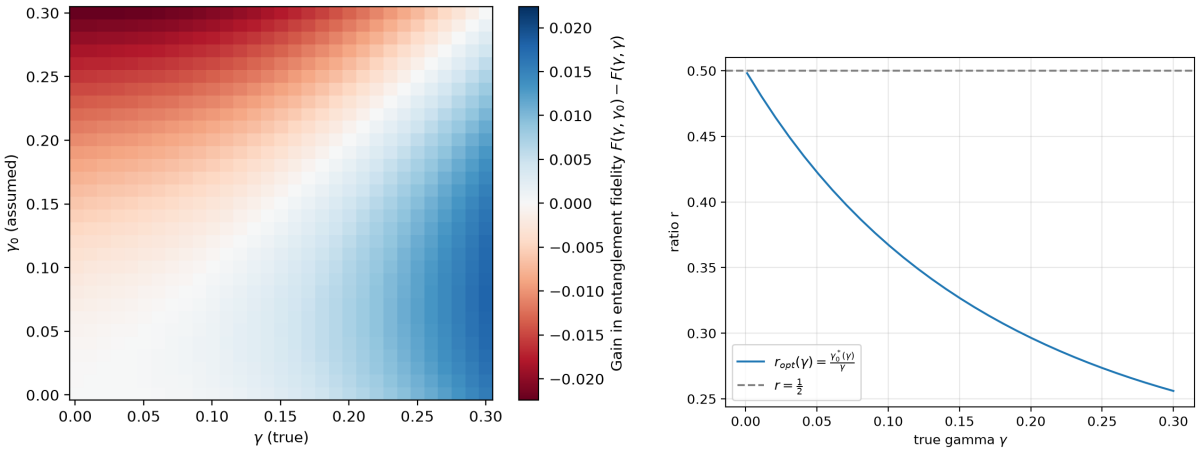
We study this question for the four-qubit code of Leung, Nielsen, Chuang and Yamamoto [3] under independent amplitude damping. Figure 1(a) shows the gain of the mismatched Petz map over the matched one ($h = \gamma$). It contains two surprises:

- *Underestimating the noise helps.* The gain is positive below the diagonal $h = \gamma$ and negative above it. The best assumed value $h^*(\gamma)$ satisfies $h^*/\gamma \approx 0.50$ at $\gamma = 10^{-3}$, and the ratio decreases to 0.26 at $\gamma = 0.3$ (Figure 1(b)).
- *Assuming no noise at all is catastrophic.* Every $h > 0$, however small, gives $1 - F = O(\gamma^2)$, whereas $h = 0$ gives $1 - F \approx \frac{3}{2}\gamma$.

This note explains both observations analytically in the weak-noise regime.

2 Mathematical setting

Noise. Amplitude damping with parameter $\gamma \in [0, 1]$ has Kraus operators $K_0(\gamma) = |0\rangle\langle 0| + \sqrt{1-\gamma}\,|1\rangle\langle 1|$ and $K_1(\gamma) = \sqrt{\gamma}\,|0\rangle\langle 1|$. On four qubits the Kraus operators are $N_j(\gamma) = K_{j_1}(\gamma) \otimes \cdots \otimes K_{j_4}(\gamma)$, labelled by bit strings $j = j_1 j_2 j_3 j_4$, where $j_i = 1$ means that qubit i decayed. We write $|j|$ for the number of decays and $\bar{j}$ for the complementary string.



(a) gain $F(\gamma, h) - F(\gamma, \gamma)$ (b) optimal ratio $h^*(\gamma)/\gamma$

Figure 1: Numerics from the notebooks `robustness.ipynb` and `petz_calibration.ipynb`; the assumed parameter h is written γ_0 on the axes. (a) Gain of the Petz map built for h over the matched map; blue means that the miscalibrated map is better. (b) The assumed parameter $h^*(\gamma)$ that maximises $F(\gamma, h)$, divided by γ. The dashed line is the weak-noise limit $\frac{1}{2}$ predicted by Theorem 1.

Code. The Leung code is spanned by $|0_L\rangle = \frac{1}{\sqrt{2}}(|0000\rangle + |1111\rangle)$ and $|1_L\rangle = \frac{1}{\sqrt{2}}(|0011\rangle + |1100\rangle)$. Its encoding isometry is $V = |0_L\rangle\langle 0| + |1_L\rangle\langle 1|$ and its code projector is $\Pi = VV^\dagger$. The code is supported on $\mathcal{C} = \mathcal{C}_0 \cup \mathcal{C}_1$, where $\mathcal{C}_0 = \{0000, 1111\}$ and $\mathcal{C}_1 = \{0011, 1100\}$. The encoded noise channel Φ_γ has the 16 Kraus operators $E_j(\gamma) = N_j(\gamma)V \in \mathbb{C}^{16\times 2}$.

Mismatched Petz map. The Petz map built for the assumed parameter h, with reference state $I_2/2$, has the Kraus operators

$$R_k(h) = \frac{1}{\sqrt{2}}\, E_k(h)^\dagger\, \tau_h^{-1/2}, \qquad \tau_h = \Phi_h\Big(\frac{I_2}{2}\Big) = \frac{1}{2}\sum_j E_j(h)E_j(h)^\dagger. \tag{1}$$

For $0 < h < 1$, τ_h is invertible (Section 4). Hence $\sum_k R_k^\dagger R_k = \tau_h^{-1/2}\tau_h\tau_h^{-1/2} = I_{16}$, and the map is trace preserving without any completion. At $h = 0$, by contrast, $\tau_0 = \Pi/2$ is singular. With a pseudo-inverse and a completion on $\ker\tau_0$, as in the code, the map becomes the baseline recovery $\mathcal{R}_0(X) = V^\dagger XV + \mathrm{Tr}[(I - \Pi)X]\, I_2/2$.

Fidelity. The entanglement fidelity of a qubit channel with Kraus operators L_i is $\frac{1}{4}\sum_i |\,\mathrm{Tr}\, L_i|^2$. The composition $\mathcal{R}_h \circ \Phi_\gamma$ has the 256 Kraus operators $R_k(h)E_j(\gamma)$, so

$$F(\gamma, h) = \frac{1}{8}\sum_{k,j} T_{kj}^2, \qquad T_{kj}(\gamma, h) = \mathrm{Tr}\left(E_k(h)^\dagger\, \tau_h^{-1/2}\, E_j(\gamma)\right) \in \mathbb{R}. \tag{2}$$

3 Main result

Theorem 1. *Fix $r > 0$ and set $h = r\gamma$. As $\gamma \to 0^+$,*

$$\boxed{F(\gamma, r\gamma) = 1 - c(r)\,\gamma^2 + O(\gamma^3), \qquad c(r) = \frac{7}{4} - \kappa\, r(1 - r), \qquad \kappa = 1 - \frac{1}{\sqrt{2}} \approx 0.293.} \tag{3}$$

The remainder is uniform for r in compact subsets of $(0, \infty)$.

Since $r(1-r) = \frac{1}{4} - (r - \frac{1}{2})^2$, the coefficient can also be written $c(r) = \frac{3}{2} + \frac{\sqrt{2}}{8} + \kappa\,(r - \frac{1}{2})^2$. This form gives the consequences for calibration.

Corollary 2. *Let c and κ be as in (3).*

(a) Matched map: $1 - F(\gamma, \gamma) = \frac{7}{4}\gamma^2 + O(\gamma^3)$.

(b) Best calibration: *c is minimal at $r = \frac{1}{2}$, so the best Petz map is built for half the true noise, with $c(\frac{1}{2}) = \frac{3}{2} + \frac{\sqrt{2}}{8} \approx 1.677$. Its gain over the matched map is $F(\gamma, \frac{\gamma}{2}) - F(\gamma, \gamma) = \frac{2-\sqrt{2}}{8}\gamma^2 + O(\gamma^3) \approx 0.073\,\gamma^2$.*

(c) Sign of the error: *$c(r) < c(1)$ if and only if $0 < r < 1$. At leading order, any underestimate of the noise helps and any overestimate hurts, which is the sign pattern of Figure 1(a).*

(d) Optimal ratio: *if $0 < r_1 < \frac{1}{2} < r_2$, every maximiser $r^*(\gamma)$ of $r \mapsto F(\gamma, r\gamma)$ on $[r_1, r_2]$ satisfies $|r^*(\gamma) - \frac{1}{2}| = O(\sqrt{\gamma})$.*

Proof. Parts (a)–(c) follow from the two forms of c. For (d), let $f_\gamma(r) = (1 - F(\gamma, r\gamma))/\gamma^2$. The uniform remainder gives $|f_\gamma(r) - c(r)| \le C\gamma$ on $[r_1, r_2]$. Since r^* minimises f_γ, we get $\kappa(r^* - \frac{1}{2})^2 = c(r^*) - c(\frac{1}{2}) \le f_\gamma(r^*) - f_\gamma(\frac{1}{2}) + 2C\gamma \le 2C\gamma$. □

Note also that $\lim_{r\to 0^+} c(r) = c(1) = \frac{7}{4}$. At this order, a Petz map built for a vanishingly small but non-zero noise is exactly as good as the matched one.

4 Proof outline

The proof is a direct computation, organised so that most of the 256 traces in (2) never need to be computed. The detailed matrices are collected in the appendix. All formulas below were checked symbolically and against the numerical code.

Step 1: each Kraus column is a single basis vector, except for no decay

A decay pattern j acts on a basis state by turning the decayed 1's into 0's:

$$N_j(\gamma)|x\rangle = \gamma^{|j|/2}(1-\gamma)^{(|x|-|j|)/2}|x - j\rangle \quad \text{if } j \preceq x, \qquad N_j(\gamma)|x\rangle = 0 \quad \text{otherwise,}$$

where $j \preceq x$ means that x has a 1 wherever j does. Applied to the codewords, this gives the two columns of E_j:

$$\sqrt{2}\,E_j(\gamma)|0\rangle = \delta_{j,0000}|0000\rangle + \gamma^{|j|/2}(1-\gamma)^{(4-|j|)/2}|\bar{j}\rangle, \tag{4}$$

$$\sqrt{2}\,E_j(\gamma)|1\rangle = \gamma^{|j|/2}(1-\gamma)^{(2-|j|)/2}(\mathbf{1}[j \preceq 0011]|0011 - j\rangle + \mathbf{1}[j \preceq 1100]|1100 - j\rangle). \tag{5}$$

Table 4 in the appendix lists all 32 columns. Two features matter:

- Every column is a multiple of a *single* basis vector, except the two columns of the no-decay operator E_{0000}, which lie in $\operatorname{span}\mathcal{C}_0$ and $\operatorname{span}\mathcal{C}_1$.
- A single decay ($|j| = 1$) sends $|0_L\rangle$ to $|\bar{j}\rangle$, of weight 3, and $|1_L\rangle$ to $|j'\rangle$, the other weight-one string of the same pair $\{0001, 0010\}$ or $\{0100, 1000\}$. The four single decays therefore leave eight distinct, mutually orthogonal syndromes.

Step 2: τ_h is block diagonal

By (1), $\langle x|\tau_h|y\rangle = \frac{1}{2}\sum_j \left(N_j(h)^\dagger|x\rangle\right)^\dagger \Pi \left(N_j(h)^\dagger|y\rangle\right)$, and $N_j^\dagger|x\rangle \propto |x+j\rangle$ re-excites the decayed qubits. This vector is zero unless x has a 0 wherever j has a 1. The projector Π only couples two strings of the same $\mathcal{C}_\ell$, and these two strings are complements of each other. For $x \neq y$ this forces $x+j$ and $y+j$ to be complementary. That is only possible if $j = 0000$ and $y = \bar{x}$ with $\{x, \bar{x}\} = \mathcal{C}_0$ or $\mathcal{C}_1$. Hence, after reordering the basis,

$$\tau_h \cong B_0(h) \oplus B_1(h) \oplus \lambda_3(h) I_4 \oplus \lambda_1(h) I_4 \oplus \lambda_2(h) I_4. \tag{6}$$

Here B_0 acts on $\mathcal{H}_0 = \mathrm{span}\{|0000\rangle, |1111\rangle\}$, B_1 acts on $\mathcal{H}_1 = \mathrm{span}\{|0011\rangle, |1100\rangle\}$, and the twelve remaining basis states are eigenvectors:

$$B_0 = \frac{1}{4}\begin{pmatrix} (1+h^2)^2 & (1-h)^2 \\ (1-h)^2 & (1-h)^4 \end{pmatrix}, \qquad B_1 = \frac{(1-h)^2}{4}\begin{pmatrix} 1+h^2 & 1 \\ 1 & 1+h^2 \end{pmatrix},$$

$$\lambda_1 = \tfrac{1}{4}h(1-h)(1+h^2) \;\; (|x|=1), \qquad \lambda_2 = \tfrac{1}{4}h^2(1-h)^2 \;\; (x \in \mathcal{W}), \qquad \lambda_3 = \tfrac{1}{4}h(1-h)^3 \;\; (|x|=3),$$

where $\mathcal{W} = \{0101, 0110, 1001, 1010\}$ are the four weight-two strings outside $\mathcal{C}$. All blocks are positive definite for $0 < h < 1$, so τ_h is invertible. It is, however, badly conditioned: six of its eigenvalues are of order $h^2/4$. The inverse square root $\tau_h^{-1/2}$ has the same block structure, with explicit blocks given in Appendix B.

Step 3: why only 20 of the 256 traces survive

Write $\tilde{e}_{j,\ell} = \sqrt{2}\, E_j|\ell\rangle$ for the columns. A 2×2 trace is the sum of the two diagonal entries, so

$$T_{kj} = \frac{1}{2}\sum_{\ell \in \{0,1\}} \tilde{e}_{k,\ell}(h)^T \, \tau_h^{-1/2} \, \tilde{e}_{j,\ell}(\gamma). \tag{7}$$

Two facts now decide which traces can be non-zero:

1. the trace only pairs *the same* logical column ℓ of $E_k(h)$ and $E_j(\gamma)$;
2. by (6), $\tau_h^{-1/2}$ does not mix the 14 blocks $\mathcal{H}_0$, $\mathcal{H}_1$ and $\mathbb{C}|x\rangle$ $(x \notin \mathcal{C})$.

So $T_{kj} \neq 0$ requires some ℓ for which column ℓ of E_k and column ℓ of E_j lie in the same block. By Step 1, every non-zero column lies in exactly one block, as listed in Table 1.

block	column $\ell = 0$ of E_j for	column $\ell = 1$ of E_j for
$\mathcal{H}_0$	$j = 0000,\ 1111$	$j = 0011,\ 1100$
$\mathcal{H}_1$	$j = 0011,\ 1100$	$j = 0000$
$\mathbb{C}\|x\rangle$, $\|x\| = 3$	$j = \bar{x}$	—
$\mathbb{C}\|x\rangle$, $\|x\| = 1$	$j = \bar{x}$	$j = x'$
$\mathbb{C}\|x\rangle$, $x \in \mathcal{W}$	$j = \bar{x}$	—

Table 1: Which columns land in which block of $\tau_h^{-1/2}$. The column $\ell = 1$ vanishes for $j \in \mathcal{W}$, $|j| = 3$ and $j = 1111$.

Reading the table row by row:

- a block $\mathbb{C}|x\rangle$ receives at most one column for each ℓ, so it only produces diagonal traces T_{jj}; this gives the 12 traces with $j \notin \mathcal{C}$;

- $\mathcal{H}_0$ with $\ell = 0$ couples 0000 and 1111, which gives the 4 traces in $\mathcal{C}_0 \times \mathcal{C}_0$;
- $\mathcal{H}_1$ with $\ell = 0$ and $\mathcal{H}_0$ with $\ell = 1$ couple 0011 and 1100, which gives the 4 traces in $\mathcal{C}_1 \times \mathcal{C}_1$;
- a mixed pair such as $(0000, 0011)$ never meets: for $\ell = 0$ its columns lie in $\mathcal{H}_0$ and $\mathcal{H}_1$, and for $\ell = 1$ in $\mathcal{H}_1$ and $\mathcal{H}_0$.

In total $12 + 4 + 4 = 20$ traces survive, and the other 236 vanish identically.

Key idea. The Petz operator $R_k(h)$ only responds to the syndrome left by the error k. Different decay patterns leave orthogonal syndromes, so R_k ignores E_j for $j \neq k$. The only exceptions are errors that send the state back into the code support $\mathcal{C}$: the no-decay operator and the double decays 0011, 1100, 1111. They form the two small groups $\mathcal{C}_0$ and $\mathcal{C}_1$.

Step 4: the twenty traces, and why single decays are corrected for every $h > 0$

Each surviving trace is a product of a few coefficients from (4)–(5) with one entry of $\tau_h^{-1/2}$. For a single decay ($|j| = 1$) the two columns lie in the blocks $\mathbb{C}|\bar{j}\rangle$ and $\mathbb{C}|j'\rangle$, where $\tau_h^{-1/2}$ is multiplication by $\lambda_3(h)^{-1/2}$ and by $\lambda_1(h)^{-1/2}$. The small coefficients $\sqrt{h}$ of $E_j(h)$ cancel against the large factors $1/\sqrt{h}$ of $\tau_h^{-1/2}$:

$$\tfrac{1}{2}\, h^{1/2}(1-h)^{3/2}\, \lambda_3(h)^{-1/2} = 1, \qquad \tfrac{1}{2}\, (h(1-h))^{1/2} \lambda_1(h)^{-1/2} = (1+h^2)^{-1/2}. \tag{8}$$

Therefore

$$T_{(1)} := T_{jj} = \sqrt{\gamma(1-\gamma)}\left[(1-\gamma) + \frac{1}{\sqrt{1+h^2}}\right] \qquad (|j| = 1). \tag{9}$$

At leading order this does not depend on h at all: on the syndromes of a single decay, $R_j(h)$ acts as the ideal correction, whatever $h > 0$. The other diagonal traces are $T_{(\mathcal{W})} = \gamma(1-\gamma)$ for $j \in \mathcal{W}$ and $T_{(3)} = h\gamma^{3/2}(1-\gamma)^{1/2}(1+h^2)^{-1/2}$ for $|j| = 3$. The no-decay trace involves both $B_0^{-1/2}$ and $B_1^{-1/2}$:

$$T_{0000,0000} = \frac{1 + (1-\gamma)^2 u}{t} + \frac{2(1-\gamma)}{w}, \tag{10}$$

where $w = \sqrt{2+h^2}$, $u = (1-h)^2 + hw$ and $t = \sqrt{1+u^2}$. The seven remaining traces, in $\mathcal{C}_0 \times \mathcal{C}_0$ and $\mathcal{C}_1 \times \mathcal{C}_1$, are listed in Appendix C. At order γ^2, only the pair $T_{0011,0011} = T_{1100,1100} = \gamma + O(\gamma^2)$ matters. With multiplicities, (2) becomes

$$8F(\gamma, h) = T_{0000,0000}^2 + 4T_{(1)}^2 + 4T_{(\mathcal{W})}^2 + 2T_{0011,0011}^2 + \rho(\gamma, h), \tag{11}$$

where ρ collects the squares of the nine other traces (Appendix C). It is of order γ^4 when $h = r\gamma$.

Step 5: why the first-order terms cancel

Put $h = r\gamma$ and expand. From (9) and (10),

$$T_{0000,0000}^2 = 8 - 16\gamma + O(\gamma^2), \qquad 4T_{(1)}^2 = 16\gamma + O(\gamma^2),$$

and all other terms of (11) are $O(\gamma^2)$. The two first-order terms cancel exactly, for every $r > 0$.

Key idea. Even when no decay happens, the state loses amplitude: E_{0000} shrinks $|1111\rangle$, $|0011\rangle$ and $|1100\rangle$. So the no-decay branch carries only the fraction $\frac{1}{2}\operatorname{Tr} E_{0000}^\dagger E_{0000} = 1 - 2\gamma + O(\gamma^2)$ of the total weight, and the missing 2γ sits in the four single-decay branches. By (8), the Petz map returns all of it to the code space, for any $h > 0$. Nothing is lost at first order, so r can only appear at order γ^2.

Step 6: where the coefficient $c(r)$ comes from

At order γ^2, the r-dependence comes only from $T_{0000,0000}$. Up to $O(h^3)$, the quantities in (10) expand as

$$\frac{1}{t} = \frac{1}{\sqrt{2}}\Big[1 + \frac{2-\sqrt{2}}{2}\,h + \frac{1-2\sqrt{2}}{4}\,h^2\Big], \qquad \frac{u}{t} = \frac{1}{\sqrt{2}}\Big[1 - \frac{2-\sqrt{2}}{2}\,h + \frac{6\sqrt{2}-7}{4}\,h^2\Big],$$

$$\frac{2}{w} = \sqrt{2} - \frac{\sqrt{2}}{4}\,h^2.$$

We then split the first term of (10) as

$$\frac{1+(1-\gamma)^2 u}{t} = \Big(\frac{1}{t} + \frac{u}{t}\Big) - (2\gamma - \gamma^2)\,\frac{u}{t},$$

where the terms linear in h cancel inside the bracket. This gives

$$T_{0000,0000} = 2\sqrt{2} - 2\sqrt{2}\,\gamma + \Big[\frac{\sqrt{2}}{2} + \underbrace{(\sqrt{2}-1)\,r}_{\text{cross term } \gamma\cdot h} \underbrace{-\,(\sqrt{2}-1)\,r^2}_{\text{terms in } h^2}\Big]\gamma^2 + O(\gamma^3). \tag{12}$$

The term linear in r is a cross term: it pairs the true deformation $(1-\gamma)^2 \approx 1-2\gamma$ of the codeword $|0_L\rangle$ with the first-order dependence of the map on h, and it rewards building the map for a larger h. The term in r^2 is a pure h^2 effect of the map itself and penalises a large h. The two coefficients are equal and opposite. This produces $r(1-r)$, which is maximal at $r = \frac{1}{2}$ and takes the same value as $r \to 0^+$ and at $r = 1$. Squaring (12) and collecting all contributions to (11) gives Table 2.

contribution	γ^0	γ^1	γ^2
$T^2_{0000,0000}$	8	-16	$12 + (8-4\sqrt{2})\,r(1-r)$
$4T^2_{(1)}$	0	$+16$	-32
$4T^2_{(\mathcal{W})}$	0	0	$+4$
$2T^2_{0011,0011}$	0	0	$+2$
ρ	0	0	0
$8F(\gamma, r\gamma)$	8	0	$-14 + (8-4\sqrt{2})\,r(1-r)$

Table 2: Coefficients of γ^0, γ^1 and γ^2 in (11) at $h = r\gamma$.

Dividing by 8, and using $\frac{8-4\sqrt{2}}{8} = 1 - \frac{1}{\sqrt{2}} = \kappa$, gives Theorem 1. As for the remainder: the factors $\sqrt{\gamma}$ in $T_{(1)}$ and $T_{(3)}$ only enter through their squares, so the right-hand side of (11) is a smooth function of (γ, r) near $\gamma = 0$. Taylor's theorem then bounds the remainder by $O(\gamma^3)$, uniformly for r in compact sets.

5 Numerical verification and discussion

Leading coefficient. Table 3 compares $(1 - F(\gamma, r\gamma))/\gamma^2$, computed with the function `mismatched_petz_fidelity`, with $c(r)$. The difference shrinks linearly in γ, as expected from the $O(\gamma^3)$ remainder. The twenty traces of Step 4 agree with a direct numerical evaluation to about 10^{-15}, and the other 236 vanish. Below $\gamma \approx 10^{-4}$ the numerical code loses accuracy, because of the eigenvalues of order h^2 in τ_h. The closed form (11), evaluated in high precision, keeps converging to $c(r)$.

r	$\gamma = 10^{-1}$	$\gamma = 10^{-2}$	$\gamma = 10^{-3}$	$c(r)$
1/4	1.6298	1.6889	1.6945	1.6951
1/2	1.6311	1.6728	1.6764	1.6768
1	1.7624	1.7528	1.7503	1.7500
2	2.5087	2.3580	2.3380	2.3358

Table 3: $(1 - F(\gamma, r\gamma))/\gamma^2$ from the code, compared with the coefficient $c(r)$ of Theorem 1.

Optimal calibration. The optimiser `optimal_petz_calibration` gives $h^*/\gamma = 0.497$ at $\gamma = 10^{-3}$ and 0.482 at $\gamma = 10^{-2}$, approaching $\frac{1}{2}$ (Figure 1(b)). At $\gamma = 10^{-3}$ the gain over the matched map is 7.39×10^{-8}, against $\frac{2-\sqrt{2}}{8}\gamma^2 = 7.32 \times 10^{-8}$. At larger γ higher orders take over: the optimal ratio drops to 0.37 at $\gamma = 0.1$ and 0.26 at $\gamma = 0.3$, and the gain exceeds the γ^2 prediction (1.37×10^{-3} against 0.73×10^{-3} at $\gamma = 0.1$).

Comparison with optimal recovery. The SDP-optimal recovery [4] gives $(1 - F_{\text{opt}})/\gamma^2 \approx 1.250$ for $\gamma \leq 0.05$ (`data/fidelities.csv`). At weak noise, the best calibration therefore closes only $(c(1) - c(\frac{1}{2}))/(c(1) - 1.25) \approx 15\%$ of the gap between the matched Petz map and the optimum. It helps more at stronger noise: at $\gamma = 0.3$ it raises F from 0.849 to 0.867, against $F_{\text{opt}} = 0.890$, which closes 44% of the gap. The constant 1.25 is a numerical observation, not part of the theorem.

The singular point $h = 0$. Theorem 1 gives $c(0^+) = \frac{7}{4}$, yet the map used at $h = 0$ is much worse. The formulas of Step 4 are continuous at $h = 0$, so

$$F(\gamma, 0^+) = \lim_{h \to 0^+} F(\gamma, h) = 1 - \tfrac{7}{4}\gamma^2 + \tfrac{3}{4}\gamma^3 + \tfrac{1}{4}\gamma^4.$$

For the baseline map $\mathcal{R}_0$, the trace $\operatorname{Tr} V^\dagger E_j(\gamma)$ is non-zero only for $j = 0000$ and $j = 1111$, and the completion contributes $\frac{1}{8}\sum_{j,\ell} \|(I - \Pi)E_j|\ell\rangle\|^2$. This gives

$$F(\gamma, 0) = 1 - \tfrac{3}{2}\gamma + \gamma^2 - \tfrac{1}{4}\gamma^3, \qquad F(\gamma, 0^+) - F(\gamma, 0) = \tfrac{\gamma}{4}(1 - \gamma)^2(6 + \gamma).$$

The reason for the jump is the cancellation (8). For every $h > 0$, the single-decay syndromes lie in the support of τ_h, with weight of order h, and the Petz map rescales them by $h^{-1/2}$, recovering them fully. At $h = 0$ they lie outside the support. The completion then replaces these branches by the maximally mixed state, which keeps only $\frac{1}{4}$ of their weight 2γ in the fidelity, so $\frac{3}{2}\gamma$ is lost. Numerically, $F(0.1, 0) = 0.85975$ while $F(0.1, 10^{-4}) = 0.98328$, close to $F(0.1, 0^+) = 0.983275$. The limits do not commute: $\lim_{r \to 0^+} \lim_{\gamma \to 0^+} (1 - F)/\gamma^2 = \frac{7}{4}$, whereas $(1 - F(\gamma, 0))/\gamma^2 \to \infty$.

Practical message and limitations. If the damping rate is uncertain, it is better to err on the low side: at weak noise, any factor between 0 and 1 helps and the factor $\frac{1}{2}$ is optimal. The guess should never be set to zero, though. The result is asymptotic in γ, and the optimality statement holds for r in a compact interval; the numerics indicate that the global optimum also tends to $\frac{1}{2}$. The structural argument of Steps 3 and 5 (orthogonal syndromes, and single decays corrected for any $h > 0$) does not rely on the details of the Leung code and should carry over to other codes for amplitude damping.

A Kraus columns

Table 4 lists the columns of $\sqrt{2}\, E_j(\gamma)$ from (4)–(5), grouped by type of decay. For $|j| = 1$, j' is the partner defined in Step 1.

j	$\sqrt{2}\,E_j(\gamma)\lvert 0\rangle$	$\sqrt{2}\,E_j(\gamma)\lvert 1\rangle$
0000	$\lvert 0000\rangle + (1-\gamma)^2\lvert 1111\rangle$	$(1-\gamma)(\lvert 0011\rangle + \lvert 1100\rangle)$
$\lvert j\rvert = 1$	$\gamma^{1/2}(1-\gamma)^{3/2}\lvert\bar{j}\rangle$	$\gamma^{1/2}(1-\gamma)^{1/2}\lvert j'\rangle$
0011, 1100	$\gamma(1-\gamma)\lvert\bar{j}\rangle$	$\gamma\lvert 0000\rangle$
$j \in \mathcal{W}$	$\gamma(1-\gamma)\lvert\bar{j}\rangle$	0
$\lvert j\rvert = 3$	$\gamma^{3/2}(1-\gamma)^{1/2}\lvert\bar{j}\rangle$	0
1111	$\gamma^2\lvert 0000\rangle$	0

Table 4: The columns of $\sqrt{2}\,E_j(\gamma)$ in the computational basis.

B The blocks of $\tau_h^{-1/2}$

The $\mathcal{H}_1$ block. B_1 has the eigenvectors $\frac{1}{\sqrt{2}}(1,\pm 1)$, with eigenvalues

$$\mu_+ = \tfrac{1}{4}(1-h)^2(2+h^2), \qquad \mu_- = \tfrac{1}{4}h^2(1-h)^2.$$

Hence

$$B_1^{-1/2} = \begin{pmatrix} \alpha & \beta \\ \beta & \alpha \end{pmatrix}, \qquad \alpha = \frac{1}{1-h}\Big(\frac{1}{h}+\frac{1}{w}\Big), \qquad \beta = \frac{1}{1-h}\Big(\frac{1}{w}-\frac{1}{h}\Big).$$

The $\mathcal{H}_0$ block. For a 2×2 positive definite matrix M, the Cayley–Hamilton theorem gives $\sqrt{M} = (M + \sqrt{\det M}\,I)/\sqrt{\operatorname{Tr} M + 2\sqrt{\det M}}$. Applied to $M = 4B_0$, for which $\sqrt{\det M} = (1-h)^2hw =: s$ and $\operatorname{Tr} M + 2s = 1 + u^2 = t^2$, this gives

$$B_0^{-1/2} = \frac{2}{s\,t}\begin{pmatrix} (1-h)^4+s & -(1-h)^2 \\ -(1-h)^2 & (1+h^2)^2+s \end{pmatrix} = \begin{pmatrix} a_0 & c_0 \\ c_0 & d_0 \end{pmatrix},$$

with $a_0 = \dfrac{2u}{hwt}$, $c_0 = -\dfrac{2}{hwt}$ and $d_0 = \dfrac{2(1+hwu)}{(1-h)^2hwt}$.

The other blocks. On $\mathbb{C}\lvert x\rangle$ with $x \notin \mathcal{C}$, $\tau_h^{-1/2}$ is multiplication by $\lambda_1^{-1/2}$, $\lambda_2^{-1/2}$ or $\lambda_3^{-1/2}$.

C The remaining traces

Each trace follows from (7) and Table 1. For instance, the columns of E_{0000} give, with the identity $u - (1-h)^2 = hw$,

$$\begin{aligned} T_{0000,0000} &= \tfrac{1}{2}\big(1,(1-h)^2\big)B_0^{-1/2}\big(1,(1-\gamma)^2\big)^T + \tfrac{1}{2}(1-h)(1-\gamma)\,(1,1)\,B_1^{-1/2}(1,1)^T \\ &= \frac{1+(1-\gamma)^2u}{t} + \frac{2(1-\gamma)}{w}. \end{aligned}$$

In the same way,

$$T_{0000,1111} = \frac{\gamma^2}{t}, \qquad T_{1111,0000} = \frac{h\big(u-(1-\gamma)^2\big)}{wt}, \qquad T_{1111,1111} = \frac{h\gamma^2u}{wt},$$

$$T_{0011,0011} = T_{1100,1100} = \frac{\gamma}{2}\Big[(1-\gamma)\Big(1+\frac{h}{w}\Big) + \frac{2u}{wt}\Big],$$

$$T_{0011,1100} = T_{1100,0011} = \frac{\gamma}{2}\Big[(1-\gamma)\Big(\frac{h}{w}-1\Big) + \frac{2u}{wt}\Big].$$

The remainder in (11) is

$$\rho = T^2_{0000,1111} + T^2_{1111,0000} + T^2_{1111,1111} + 2T^2_{0011,1100} + 4T^2_{(3)}.$$

At $h = r\gamma$ every term of ρ is of order γ^4 or smaller, since $T_{0011,1100}$, $T_{0000,1111}$ and $T_{1111,0000}$ are $O(\gamma^2)$, $T_{1111,1111} = O(\gamma^3)$ and $T_{(3)} = O(\gamma^{5/2})$.

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